Architectures, Services and Attacks

Plenary

Enterprise Cyber Security

Today

Putting last week's lecture into action

- Stay in your **groups**. One scenario, **Sammet Housing Association**, and three tasks that use what we covered last week.
- For each task: you work in your **group**, then we **debrief** as a class.
- Everyone writes their own answer, on paper or a device. Short, specific answers are the aim. The scenario and each task stay on screen.
- The material is **Parts 1 and 2** of last week's lecture: the architectural eras, and cloud service and deployment models.
- The slides are yours afterwards: every task is followed by the debrief.

Before we start

Four things to remember

- Read the **“In your answer”** sentence first. It lists what a good answer covers.
- Knowledge parts: be **precise**. Judgement parts: **identify** a point, **apply** it to the scenario, and **justify** it.
- A concern that could be written about any organisation, such as “data could be stolen”, says little. Use **the scenario**.
- Task 3 asks for the most, so expect to write the most there.

SET-UP
ERAS
SERVICE MODELS
DEPLOYMENT MODELS
CHECK IT

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Sammet
Housing
Association

Sammet Housing Association

A fictional example

- A housing association with about 6,000 homes in the west of Scotland.
- **Rent accounting** runs on an in-house application written in COBOL in the 1980s. It processes payments in overnight batch runs on a central computer at head office. Only one programmer, who retires next year, fully understands it.
- Twelve **neighbourhood offices** each choose their own software for local work, and keep rent arrears spreadsheets on office PCs.

Servers and suppliers

Who connects, and who runs what

- The offices' desktop software also connects to a single tenancy database server, `tn01`, at head office. It holds tenant names, addresses, rent accounts and notes on vulnerable tenants, such as those who need adapted homes.
- Repairs are carried out by a contractor, **Wang Property Services**. Wang staff log in remotely over the Internet, from anywhere, to update repair jobs.
- Tenants report repairs and pay rent through a portal and mobile app run by an external supplier, **Dean Cloud Services**.

SET-UP

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Task 1:
Eras

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Task 2:
Service Models

Next year: a new repairs system

Dean offers two ways to host it

- Sammet will replace its repairs system. Dean Cloud Services offers two options for hosting it.
- **Option A**: Dean provides virtual machines. Sammet installs and runs the repairs software and its database on them. Dean looks after the physical servers and the virtualisation software.
- **Option B**: Dean provides a managed platform. Sammet deploys the repairs software on it, decides its settings, and uses Dean's own database service. Dean looks after everything beneath the repairs software.

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Task 3: Deployment Models

The trustees disagree

Five associations, one platform

- Sammet is one of **five housing associations** in the region considering sharing one repairs platform, on Option B.
- One trustee wants Dean's **public cloud**, which is the cheapest. Dean runs it in data centres across Europe and North America, and may move workloads between them to balance demand.
- Sammet's funding agreement requires **tenant data to stay in the UK**.
- Dean has also offered a cloud **reserved for the five associations alone**, with the same platform, at a higher price.

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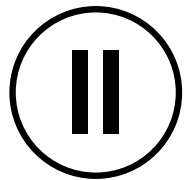
Check It Yourself

Putting it into action

Take-aways from today

- Name the era **and** the feature that shows it, then state a concern in **the scenario's terms**.
- Know the **customer's side of the line**: access and your own settings stay with you in every service model.
- For deployment models, ask the **three questions**: where, who controls the virtualisation, who configures and maintains.
- Weigh **both sides**, then recommend and justify. More than one answer works.
- What the logs show depends on the **service model**. Agree it before an incident.
- **Practice**: write Task 3 in full in 15 minutes, then check it against the debrief.

Any questions?



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